0 1 2 3 4

arr = [100, 30, 55, 90, 70]

mini = 100, 30, 30, 30, 30

maxi = 100, 100, 100, 100, 100

buy = 30 25, 60, 40

sell = 90

profit = 90 - 30 = 60

profit = max(arr[i] - mini, profit)

- mini = INT_MAX;

  maxi = INT_MIN;

- mini denotes minimum value in array till index i.
- maxi denotes maximum value in array till index i.

- <u>Subarray</u> :- It is a contiguous range of array elements from index location i till i+K.

[arr[i], arr[i+K]]

$$arr = [\overset{0}{10}, \overset{1}{20}, \overset{2}{30}, \overset{3}{40}, \overset{4}{50}]$$

① $arr(0) = 10$, $arr(1) = 20$, $arr(2) = 30$

$arr(3) = 40$, $arr(4) = 50$

① (10, 20), (20, 30), (30, 40), (40, 50)

① (10, 20, 30), (20, 30, 40), (30, 40, 50)

① (10, 20, 30, 40), (20, 30, 40, 50)

① (10, 20, 30, 40, 50).

- <u>HashMap</u>:- It is a data structure which is used to store key-value pair. It is used for efficient data retrieval.

0	1	2	3	4
10	20	30	40	50

Key → value

map[10] = 0 map(40) = 3

map(20) = 1 map(50) = 4

map(30) = 2

map(key, value)

Any data Type Any data Type

0 1 2 3 4 5

1, 4, 20, 3, 10, 5

curr. sum = 1, 5, 25, 28, 38, 43

target sum = 25

map[1] = 0

map[5] = 1 → i+1

map[25] = 2

map[28] = 3

map[38] = 4

map(43) = 5

starting point of subarray

38 − 33 = 5

map[0] = −1

i+1 = 0